

Combined Mode Fracture Toughness Measurement by the Disk Test

HIDEO AWAJI

Assistant Professor,
Department of Mechanical Engineering,
Fukushima Technical College,
Iwaki, Japan

SENNOSUKE SATO

Professor,
Department of Mechanical Engineering,
Faculty of Engineering,
The University of Ibaraki,
Hitachi, Japan

The disk test in which a circular specimen with an internal crack is subjected to diametral compression is used to investigate combined mode I and mode II fracture. The stress intensity factors in the disk test are calculated numerically by means of the boundary collocation procedure and the dislocation method. Special care was taken to analyze the effect of the compression anvils. This method has the advantage, of allowing successive measurement of mode I, mode II and the combined mode fracture toughness under the same conditions. Some kinds of graphite, plaster and marble are examined to obtain the fracture toughness values, K_{IC} , K_{IIC} and the combined mode fracture criterion.

1 Introduction

There are three basic modes of crack extension, namely, the crack opening mode (mode I), the crack sliding model (mode II), and the antiplane shearing mode (mode III). Most investigations in linear elastic fracture mechanics have been carried out under mode I conditions in which the crack propagates perpendicularly to the loading direction and crack instability occurs when the stress intensity factor reaches a critical value. For plane strain conditions this is denoted as K_{Ic} . Only a limited number of theoretical and experimental investigations have been conducted so far on combined mode loading. However the fracture of brittle materials occurs at a shear stress approximately equal to the tensile fracture stress. Thus, it is important from an engineering point of view to study mode II fracture toughness and the combined mode fracture criterion. For a cracked component under combined mode loading, the following three criteria of fracture have been advanced, namely, the maximum stress criterion [1],¹ the strain energy density factor criterion [2, 3] and the maximum energy release rate criterion [4]. However, damage effects due to plastic deformation [5] may differ in mode I and mode II loading. For this reason experiments are needed to evaluate K_{IIC} and the values of K_{Ic} and K_{IIC} for combined mode loading.

Recently, some experimental investigations on combined mode fracture toughness have been carried out. Liu [6] showed $K_{IIC}/K_{Ic} = 1.18$ and 1.01 for 7075-T7651 and 2024-T32 aluminum alloys, by means of a shear pannel method. Shah [7] showed the combined mode I-II criterion for 4340 steel as follows;

$$\frac{K_I}{K_{Ic}} + \frac{K_{II}}{K_{IIC}} = 1 \quad (1)$$

and $K_{IIC}/K_{Ic} = 1.09$. Jones, et al. [8] proposed the compact shear specimen to investigate mode II fracture toughness and showed $K_{IIC}/K_{Ic} \doteq 1.16$ for 2024-T4 aluminum alloy. Also Pook [9] gave $K_{IIC}/K_{Ic} = 1$ for DTD 5050 aluminum alloy. These results show that K_{IIC}/K_{Ic} ratios are in the range of 1.0 ~ 1.2 for some aluminum alloys and steel. They do not agree with the values deduced from the three criteria mentioned above.

Nuismer [10] has analyzed the energy release rate in combined mode I-II. His analysis is based on the assumption that the energy release rate for the branch crack is the same as the value obtained by the following equation for the energy release rate when a crack is assumed to propagate within its own plane;

$$G = \frac{\pi(\kappa + 1)}{8\mu} (K_I^2 + K_{II}^2) \quad (2)$$

where K_I and K_{II} are stress intensity factors for the crack in question, in which $\kappa = 3 - 4\nu$ for plane strain and $\kappa = (3 - \nu)/(1 + \nu)$ for generalized plane stress. Since, after the initiation of a crack, its propagation proceeds in the plane of the branch crack which is considered to be $K_{II} = 0$, the assumption in equation (2) can be proved valid. But at the very beginning of crack initiation under combined modes, equation (2) is not valid, because a branch crack propagates in a deviated direction from the initial crack plane. With this assumption, Nuismer's analytical result is finally found to be into one and the same as that obtained by the maximum stress criterion.

This paper presents an analysis of a disk test in which a circular specimen with an internal crack is subjected to the diametral loading as shown in Fig. 1. The specimen configuration of Fig. 1 will permit any ratio of K_I and K_{II} including "pure" conditions by the appropriate choice of crack orientation, and these fracture toughnesses can be easily determined successively under various conditions. This method has several advantages as is shown in

¹Numbers in brackets designate References at end of paper.

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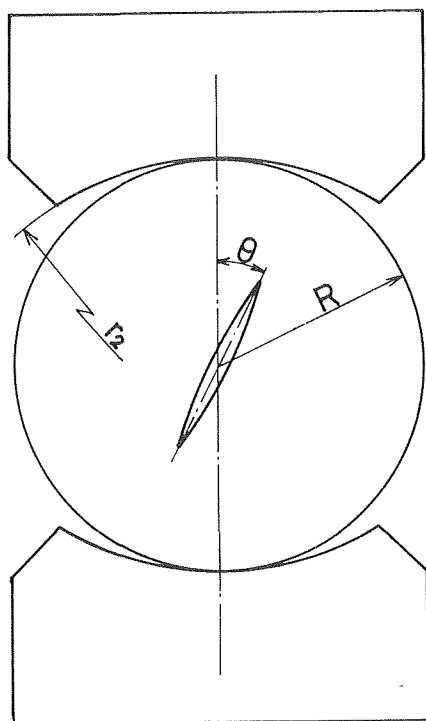


Fig. 1 Method for measuring fracture toughness in the combined mode

studies on thermal shock resistance [11], thermal shock fracture toughness [12] and neutron irradiation effects at high temperature.

In this paper, the values of stress intensity factor in the disk used for fracture toughness testing are analyzed by means of the boundary collocation method [13] and the dislocation method [14, 15]. Especially, we consider in detail the Hertzian contact stress distribution produced by the external loading at the outer boundary of the disk for the correction of a simple concentrated loading.

2 Analysis of Stress Intensity Factor in the Disk Test

2.1 The Case in Which the Disk is Subjected to a Concentrated Force. We consider the problem in which a circular disk with a central crack is subjected to the concentrated diametrically opposed forces in an inclined direction to the central crack orientation, as shown in Fig. 2. Ishida [13] solved the problem in the particular case in which the direction of concentrated load coincides with that of the central crack using boundary collocation. In this paper, we attempt to solve the problem for the general case in which the loading is inclined to the crack orienta-

tion. We use boundary collocation and the dislocation method, which involves the superposition of the following three stages of stress state similarly to Ishida's method [13].

(A) Circular region loaded by concentrated diametral forces, as shown in Fig. 2(a).

(B) Infinite region with an isolated crack subjected to distributed loadings along the crack surface which balance the stresses on the dashed line in Fig. 2(a), as shown in Fig. 2(b).

(C) Circular region containing a crack with stress-free faces and giving the assigned boundary stresses along the circular boundary which compensates the disturbing stresses which are caused by the state B, as shown in Fig. 2(c).

Solution of the stress state A is well known [16]. Analysis of the state B is solved by the dislocation method as shown in the following. The problem of an infinite region with an isolated crack subjected to distributed normal stress $p(x)$ and tangential stress $q(x)$ on its crack surface can be simulated by superposing two distributions of infinitesimal dislocation $f(\xi)$ and $g(\xi)$ between crack edges $c, -c$. From requirements for the equilibrium of each dislocation, $f(\xi)$ and $g(\xi)$ must satisfy the following equations [14];

$$\left. \begin{aligned} \int_{-c}^c \frac{f(\xi)}{\xi - x} d\xi + p(x) &= 0, \\ \int_{-c}^c \frac{g(\xi)}{x - \xi} d\xi + q(x) &= 0. \end{aligned} \right\} \quad (3)$$

Distribution functions $f(\xi)$ and $g(\xi)$ are obtained by using the inverse theorem of Muskhelishvili [17] as follows;

$$\begin{Bmatrix} f(\xi) \\ g(\xi) \end{Bmatrix} = \pm \frac{1}{\pi^2} \cdot \frac{1}{\sqrt{c^2 - \xi^2}} \int_{-c}^c \frac{\sqrt{c^2 - x^2}}{x - \xi} \begin{Bmatrix} p(x) \\ q(x) \end{Bmatrix} dx \quad (4)$$

Functions $f(\xi)$ and $g(\xi)$ can be described also in the following forms:

$$f(\xi) = \frac{F(\xi)}{\sqrt{c^2 - \xi^2}}, \quad g(\xi) = \frac{G(\xi)}{\sqrt{c^2 - \xi^2}} \quad (5)$$

where $F(\xi)$ and $G(\xi)$ are differentiable between the interval of edges $[-c, c]$. Substituting equation (5) into (4), we obtain;

$$\begin{Bmatrix} F(\xi) \\ G(\xi) \end{Bmatrix} = \pm \frac{1}{\pi^2} \int_{-c}^c \frac{\sqrt{c^2 - x^2}}{x - \xi} \begin{Bmatrix} p(x) \\ q(x) \end{Bmatrix} dx \quad (6)$$

and making a transformation of a variable such as $\xi = c \sin \phi$, we have;

$$\begin{Bmatrix} \bar{F}(\phi) \\ \bar{G}(\phi) \end{Bmatrix} = \pm \frac{1}{\pi^2} \int_{-c}^c \frac{\sqrt{c^2 - x^2}}{x - c \sin \phi} \begin{Bmatrix} p(x) \\ q(x) \end{Bmatrix} dx \quad (7)$$

where

$$\bar{F}(\phi) = F(c \sin \phi), \quad \bar{G}(\phi) = G(c \sin \phi) \quad (8)$$

Nomenclature

K_I, K_{II} = stress intensity factors
 N_I, N_{II} = nondimensional stress intensity factors
 K_{Ic}, K_{IIc} = fracture toughness
 G = energy release rate
 κ = material constant, $\kappa = 3 - 4\nu$ for plane strain, $\kappa = (3 - \nu)/(1 + \nu)$ for generalized plane stress
 ν = Poisson's ratio

μ = shear modulus
 E = Young's modulus
 $p(x), q(x)$ = distributed stresses on crack surface
 $f(\xi), g(\xi)$ = distributions of infinitesimal dislocation
 $\sigma_x, \sigma_y, \tau_{xy}$ = stresses
 R_x, R_y = resultant stresses
 σ_t = tensile strength
 $\phi(z), \Omega(z)$ = complex stress functions

$2c$ = crack length
 $2b$ = Hertzian contact width
 R = radius of disk specimen
 t = specimen thickness
 P_c = rupture load
 γ_I, γ_{II} = revised factors
 θ = crack angle (as measured from the direction of the load)

Thus the problem can be reduced to a solution of the integral equations (7). For numerical calculation of equations (7), we divide the integral interval $(-c, c)$ into m equal segments where number m was selected as 40 in the numerical calculations. Approximating the stress distribution functions $p(x)$ and $q(x)$ to linear equations $\alpha_i x + \beta_i$ in each divided segment interval $i, i+1$, and substituting them into equations (7), we can integrate those equations using integral formulas, as follows;

$$\begin{aligned} \left\{ \frac{\bar{F}(\phi)}{\bar{G}(\phi)} \right\} &= \pm \frac{1}{\pi^2} \sum_{i=1}^m \int_{D_i} \frac{\sqrt{c^2 - x^2}}{x - c \sin \phi} (\alpha_i x + \beta_i) dx \\ &= \pm \frac{1}{\pi^2} \sum_{i=1}^m \left[\sqrt{c^2 - x^2} \left(\frac{\alpha_i}{2} x + \alpha_i c \sin \phi + \beta_i \right) + \arcsin \frac{x}{c} \right. \\ &\quad \times \left(-\alpha_i c^2 \sin^2 \phi - \beta_i c \sin \phi + \frac{\alpha_i}{2} c^2 \right) \\ &\quad \left. + c \cos \phi (\alpha_i c \sin \phi + \beta_i) \log \left| \frac{x - c \sin \phi}{2c^2 - 2c \sin \phi x + 2c \cos \phi \sqrt{c^2 - x^2}} \right| \right]_{x_i}^{x_{i+1}} \end{aligned} \quad (9)$$

Stresses in an arbitrary point (x, y) are obtained from the stresses around the edge dislocation, as follows;

$$\begin{aligned}\sigma_x &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\bar{F}(\phi)(x - c \sin \phi)\{y^2 - (x - c \sin \phi)^2\} - \bar{G}(\phi)y\{3(x - c \sin \phi)^2 + y^2\}}{\{(x - c \sin \phi)^2 + y^2\}^2} d\phi \\ \sigma_y &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{-\bar{F}(\phi)(x - c \sin \phi)\{3y^2 + (x - c \sin \phi)^2\} + \bar{G}(\phi)y\{(x - c \sin \phi)^2 - y^2\}}{\{(x - c \sin \phi)^2 + y^2\}^2} d\phi \\ \tau_{xy} &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\bar{F}(\phi)y\{y^2 - (x - c \sin \phi)^2\} + \bar{G}(\phi)(x - c \sin \phi)\{(x - c \sin \phi)^2 - y^2\}}{\{(x - c \sin \phi)^2 + y^2\}^2} d\phi\end{aligned}\quad (10)$$

Next, we consider a circular region having a stress free crack and along its outer boundary subjected to resultant stresses R_x and R_y which compensate the disturbing stresses caused by the stress state B . Relations for the components of stress and resultant stress are given in term of the complex stress functions $\phi(z)$ and $\Omega(z)$ as follows:

$$\left. \begin{aligned} \sigma_x + \sigma_y &= 2[\phi(z) + \overline{\phi(x)}] \\ -\sigma_x + \sigma_y + 2i\tau_{xy} &= 2[(\bar{z} - z)\phi'(z) - \phi(z) + \bar{\Omega}(z)] \end{aligned} \right\} \quad (11)$$

$$R_y + iR_x = - \int \overline{\phi(z)} d\bar{z} - \int \bar{\Omega}(z) dz + 2iy\phi(z) \quad (12)$$

where R_x and R_y are components of the resultant stress exerted by the right-hand side upon the left-hand side across a path leading to point (x, y) from a definite point. The complex stress functions for the above stress state C are expressed as follows [13]:

$$\begin{pmatrix} \phi(z) \\ \Omega(z) \end{pmatrix} = \sum_{n=1}^{\infty} \left(L_n \frac{z^n}{\sqrt{z^2 - c^2}} \pm M_n z^n \right) \quad (13)$$

These functions do not cause any traction along the crack surface. Therefore, the unknown coefficients L_n and M_n may be determined from the boundary conditions in which $\phi(z)$ and $\Omega(z)$ equilibrate with assigned distributed stresses.

When distributed stresses on the crack surface are only the normal stress $p(x)$, the stress function $\phi_p(z)$ and $\Omega_p(z)$ become even-functional with real coefficients, as follows;

$$\begin{pmatrix} \phi_p(z) \\ \Omega_p(z) \end{pmatrix} = \sum_{n=1}^{\infty} \left(\frac{z^{2n-1}}{\sqrt{z^2 - c^2}} \pm M_{2n-1} z^{2n-2} \right) \quad (14)$$

And when distributed stress is the tangential stress $q(x)$ only, $\phi_q(z)$ and $\Omega_q(z)$ are even functions with imaginary coefficients, as follows;

$$\left\{ \begin{array}{l} \phi_q(z) \\ \Omega_q(z) \end{array} \right\} = \sum_{n=1}^{\infty} i \left(L'_{2n-1} \frac{z^{2n-1}}{\sqrt{z^2 - c^2}} \pm M'_{2n-1} z^{2n} \right) \quad (15)$$

where $\dot{L}$, $\dot{M}$, L' , and M' are real coefficients. The boundary conditions are replaced by requirements for the resultant of the applied loads. The boundary condition for the case of $p(x)$ is

$$R_{yp} + iR_{zp} = \sum_{n=1}^{\infty} \dot{L}_{2n-1} \left(-2 \int \frac{z^{2n-1}}{\sqrt{z^2 - c^2}} dz + 2iy \frac{z^{2n-1}}{\sqrt{z^2 - c^2}} \right) + \sum_{l=1}^{\infty} \dot{M}_{2n-1} 2iy z^{2n-2} \quad (16)$$

In the same manner, the equation in the case of $q(x)$ can be obtained. The supplementary compatibility relation [13],

$$\kappa \int \overline{\phi(z)} d\bar{z} - \phi \bar{\Omega}(z) dz = 0 \quad (17)$$

is satisfied inherently in this case where the stress function is even. Equation (16) is solved numerically by reducing it to linear equations for the unknown coefficients $L_{2\bar{n}-1}$ and $M_{2\bar{n}-1}$.

Stress intensity factors for stress state *B* are expressed in non-dimensional forms as follows [14];

$$\left. \begin{aligned} N_I &= \frac{K_I}{K} = \frac{\pi}{\sigma_p c} \bar{F} \left(-\frac{\pi}{2} \right), \\ N_{II} &= \frac{K_{II}}{K} = -\frac{\pi}{\sigma_p c} \bar{G} \left(-\frac{\pi}{2} \right), \end{aligned} \right\} \quad (18)$$

where $K = \sigma_p \sqrt{\pi c}$ and $\sigma_p = P/\pi R t$.

Stress intensity factors at the stress state C are expressed conclusively as follows [13].

$$(N_I - iN_{II})_{\pm c} = \pm \frac{2}{\sigma\rho} \sqrt{\frac{2}{c}} \lim_{z \rightarrow \pm c} [\sqrt{\pm(z \mp c)} \phi(z)] \quad (19)$$

2.2 The Case in Which the Hertzian Contact is Taken into Consideration.

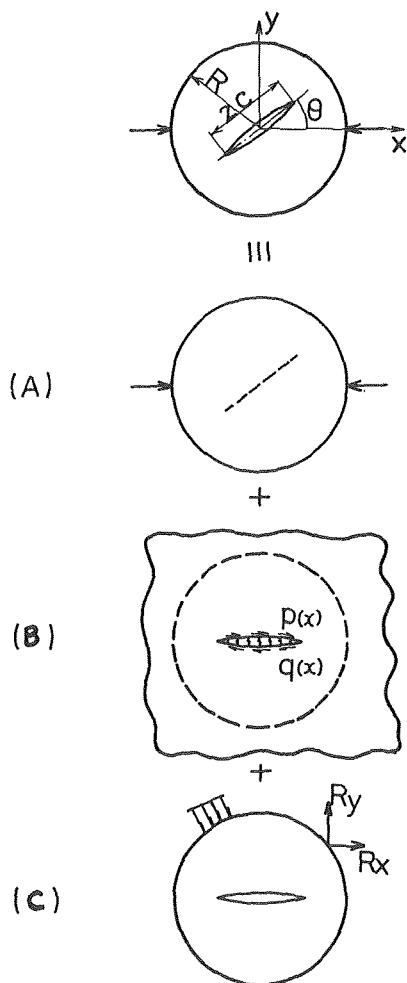


Fig. 2 Analytical approach to stress intensity factor for the disk test with central crack

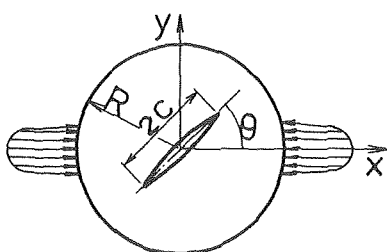


Fig. 3 The disk test with Hertzian contact pressure

disk with central crack is subjected to the Hertzian contact stress, as shown in Fig. 3. We can solve the problem by superposition of three stages of stress state in the manner shown in the preceding section, except for the stress state A. The proper problem of the state A has been dealt with already in another of our papers [18] as a special case of a circular region being loaded by distributed stress on its circular boundary.

3 Numerical Results

3.1 The Disk Subjected to a Concentrated Force. Fig. 4 shows the stress intensity factor in a circular disk with a central crack when it is subjected to diametrically concentrated forces with the crack orientation inclined to the loading axis. Comparing the plotted results in this figure with that of Ishida for c/R

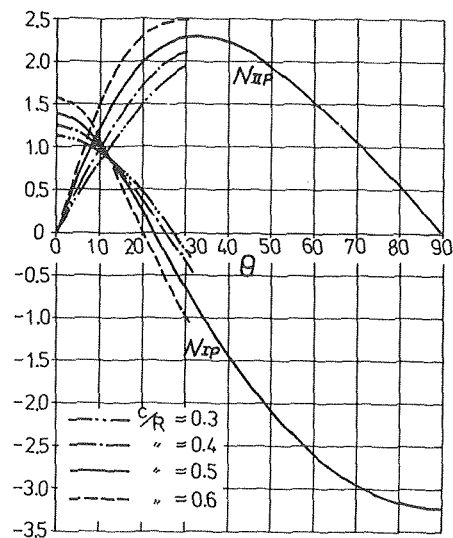


Fig. 4 Stress intensity factor of disk subjected to concentrated force

Table 1 Stress intensity factors for pure mode II and angles of crack for various crack sizes

c/R	N_{IIP}	θ_0 (deg.)
0.3	1.866	27.2
0.4	1.977	25.2
0.5	2.133	22.9
0.6	2.352	20.1

$= 0.5$ and $\theta = 0^\circ, 90^\circ$ yield values of 1.3873, -3.221 respectively, it is shown that these results agree closely with each other. And Rooke and Tweed [19] gave 1.39, -3.127 respectively, for same conditions. Their value of $\theta = 90^\circ$ is lower about 3 percent than our result. Table 1 shows the stress intensity factor for pure mode II and the angle of the inclined crack for various ratios of crack size to disk radius.

3.2 The Disk Subjected to the Hertzian Contact Stress Distribution. We consider in detail the Hertzian contact stress distribution as the external loading at the outer boundary. The nondimensional stress intensity factors N_{IH} and N_{IHH} , when the Hertzian contact width $2b$ is taken into account, are expressed by modifying those for the disk subjected to concentrated loading,

$$N_{IH} = N_{IP} - \gamma_I \left(\frac{b}{R} \right)^2$$

$$N_{IHH} = N_{IIP} - \gamma_{II} \left(\frac{b}{R} \right)^2 \quad (20)$$

where γ_I and γ_{II} are numerical factors. Fig. 5 shows values in pure mode I calculated in terms of various sizes of crack length c/R . The special case in the above equations, $b/R = 0$ is consistent with the case of concentrated force loading. As for the case of combined mode I and II stress intensity factors, the factors γ_I and γ_{II} are shown in Table 2 to 5 according to various crack lengths. Tables 6 and 7 show γ_{II} and θ_0 for pure mode II with regards to some values of crack size and contact width.

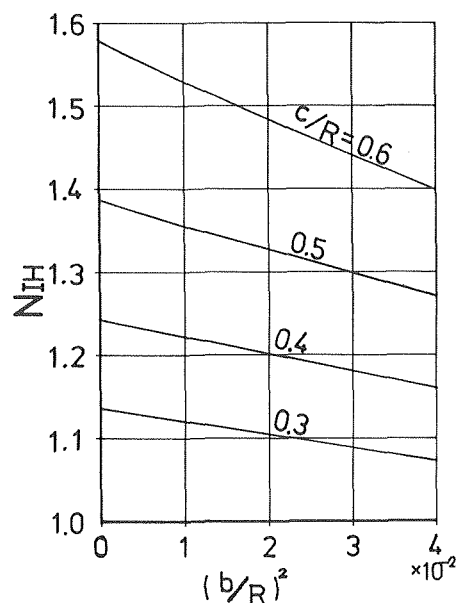


Fig. 5 Stress intensity factor in pure mode I of a disk of radius R with Hertzian contact stress over a distance $2b$

Table 2 γ_I and γ_{II} for $C/R = 0.3$

θ	MODE	$(b/R)^2 (\times 10^{-2})$			
		1	2	3	4
0	I	1.67	1.61	1.60	1.58
	II	-	-	-	-
5	I	1.60	1.54	1.53	1.52
	II	0.45	0.43	0.43	0.42
10	I	1.42	1.37	1.36	1.35
	II	0.86	0.82	0.80	0.80
15	I	1.14	1.11	1.10	1.09
	II	1.18	1.12	1.11	1.10
20	I	0.81	0.79	0.79	0.78
	II	1.41	1.34	1.32	1.31
25	I	0.46	0.46	0.46	0.46
	II	1.53	1.44	1.43	1.42
30	I	0.13	0.14	0.14	0.14
	II	1.55	1.46	1.45	1.44

Table 3 γ_I and γ_{II} for $C/R = 0.4$

θ	MODE	$(b/R)^2 (\times 10^{-2})$			
		1	2	3	4
0	I	2.19	2.11	2.09	2.07
	II	-	-	-	-
5	I	2.05	1.98	1.96	1.94
	II	0.73	0.70	0.69	0.68
10	I	1.68	1.63	1.62	1.60
	II	1.33	1.28	1.26	1.24
15	I	1.17	1.14	1.13	1.13
	II	1.73	1.66	1.64	1.62
20	I	0.63	0.62	0.62	0.62
	II	1.91	1.83	1.81	1.79
25	I	0.13	0.14	0.14	0.15
	II	1.90	1.81	1.79	1.78
30	I	-0.27	-0.25	-0.24	-0.23
	II	1.76	1.67	1.66	1.64

Table 4 γ_I and γ_{II} for $C/R = 0.5$

θ	MODE	$(b/R)^2 (\times 10^{-2})$			
		1	2	3	4
0	I	3.11	3.00	2.95	2.91
	II	-	-	-	-
5	I	2.78	2.69	2.66	2.62
	II	1.32	1.26	1.24	1.21
10	I	1.98	1.92	1.91	1.89
	II	2.26	2.17	2.13	2.09
15	I	1.00	0.98	0.98	0.99
	II	2.66	2.56	2.52	2.49
20	I	0.13	0.14	0.15	0.17
	II	2.60	2.50	2.47	2.45
25	I	-0.49	-0.46	-0.45	-0.43
	II	2.27	2.17	2.16	2.14
30	I	-0.86	-0.82	-0.80	-0.79
	II	1.84	1.74	1.74	1.73

Table 5 γ_I and γ_{II} for $C/R = 0.6$

θ	MODE	$(b/R)^2 (\times 10^{-2})$			
		1	2	3	4
0	I	4.90	4.70	4.59	4.47
	II	-	-	-	-
5	I	4.00	3.87	3.80	3.73
	II	2.72	2.60	2.51	2.43
10	I	2.05	2.01	2.02	2.01
	II	4.13	3.97	3.87	3.78
15	I	0.21	0.23	0.27	0.31
	II	4.09	3.96	3.90	3.84
20	I	-0.93	-0.89	-0.85	-0.81
	II	3.28	3.17	3.16	3.14
25	I	-1.40	-1.36	-1.33	-1.30
	II	2.35	2.26	2.26	2.26
30	I	-1.48	-1.42	-1.41	-1.39
	II	1.59	1.50	1.51	1.51

Table 6 γ_{II} for pure mode II for some values of crack size and contact width

c/R	$(b/R)^2 (\times 10^{-2})$			
	1	2	3	4
0.3	1.71	1.62	1.61	1.61
0.4	1.95	1.86	1.85	1.81
0.5	2.33	2.23	2.21	2.19
0.6	2.90	2.78	2.76	2.74

Table 7 The angle θ_0 (degrees) pure mode II for various values of crack size and contact width

c/R	$(b/R)^2 (\times 10^{-2})$			
	1	2	3	4
0.3	27.2	27.1	27.1	27.0
0.4	25.2	25.2	25.2	25.2
0.5	22.9	22.9	23.0	23.0
0.6	20.2	20.3	20.4	20.5

4 Experimental Method

4.1 Materials of Specimen. Table 8 shows the mechanical properties of specimens of graphite, plaster and marble. Graphite 7477 and SM1-24 are reactor grades having nearly isotropic properties which are obtained by molding methods. Graphite SA is an electrode graphite for the steel making furnace, which is formed by a common extruding method. The plaster specimen is made by mixing plaster of Paris and water in the ratio of 100:75 in weight, which, after being left for some days, is dried for 12 hours at 50°C. The marble used here is Italian Ondagata light marble [20].

4.2 Disk Specimen. Graphite in general is anisotropic. In this paper, the directions of cutting and loading of fracture toughness specimen were selected as follows. For graphite 7477, the fracture surface was selected to be parallel to the molding direction ($\parallel$). For SM1-24, the fracture surface was selected to be normal to the molding direction ($\perp$). Specimens of electrode graphite SA($\perp$) and SA($\parallel$) were selected to have fracture surfaces parallel to the extruding direction ($\parallel$) and normal to the extruding direction ($\perp$), respectively, specimens were cut from the central area in the cylindrical rod of manufactured materials. For the plaster specimens the hardened outer layer was scraped off with a lathe. The marble specimens were machined with a lathe and the foliation makings in the marble were oriented normal to the fracture surface.

4.3 Experimental Method of Fracture Toughness Testing. Fig. 6 shows the state of combined mode fracture toughness testing by the disk loaded by circular anvils. The circular anvils [18] are very useful when the contact area of the specimen collapses partially or if crack propagation branches off in a complicated manner.

A sharp central slit in the disk specimen was machined with a thin round saw of 13 mm diameter of 0.2 mm thickness, except

Table 8 Mechanical properties of specimens

Materials	Young's modulus E GN/m ²	Poisson's ratio ν	Tensile strength σ_t MN/m ²
SA($\perp$)	5.4	0.07	6.18
SA($\parallel$)	15.7	0.23	13.2
7477	11.8	0.20	20.7
SM1-24	9.8	—	15.5
Plaster	—	—	16.7
Marble	75.5	0.33	16.1

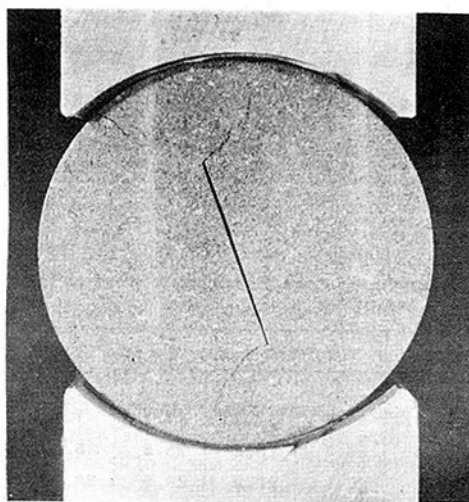


Fig. 6 Fracture appearance in the combined mode

for marble. Marble specimens were machined with a resin cutter of dental use. Then the tips of the slit were sharpened further with a razor blade with serrated edges. The slit width was about 0.3 mm for graphite and plaster and about 0.6 mm for marble. The length of slit was determined so as to be symmetrical with the center of the disk through observation with an optical comparator. The ratio of slit length and disk diameter c/R was taken to be approximately 0.5. The fracture toughness value was determined using the stress intensity factor N_{IH} and N_{III} of equations (20) corresponding to the values of γ_I , γ_{II} , and b/R . The width of the contact area was measured directly tracing the length which appeared on a thin paper by inserting a black ribbon of a lineprinter between the specimen and anvils. The loading rate on $K_{I,IIc}$ per second was around 0.01 MPa m^{1/2}/s.

5 Experimental Results and Discussions

5.1 Availability of the Disk Test. First, we examined the effects of specimen thickness on fracture toughness using graphite SA($\perp$), and obtained the fact that mode I fracture toughness K_{Ic} was independent of specimen thickness in the range of 3 to 8 mm. This agrees with the results of Yahr and Valachovic [21]. In this study, therefore, the thickness of the disk was selected as about 4 mm.

The compact tensile testing [22] has been used widely for the measurement of mode I fracture toughness. For a porous material as graphite, however, crack initiation requires less energy than crack propagation [23]. It is, therefore, difficult to obtain accurate fracture toughness values by using the compact tensile specimen, especially for electrode graphite which has large inherent defects. According to microscopic observations of crack initiation in the disk specimen, a slight deformation at the tip of the artificial crack occurs at first and the initiation shows little change of curvature on a recording chart just before propagation. Then the unstable crack propagates to final fracture. These aspects vary apparently with compact tensile specimens. With a specimen of an isotropic reactor graphite 7477 with fine structure, the disk specimen and the compact tensile specimen were compared. Direction of fracture surface was selected as normal to the molding pressure, ($\perp$). Load-displacement curves in the disk test and the compact tensile test are

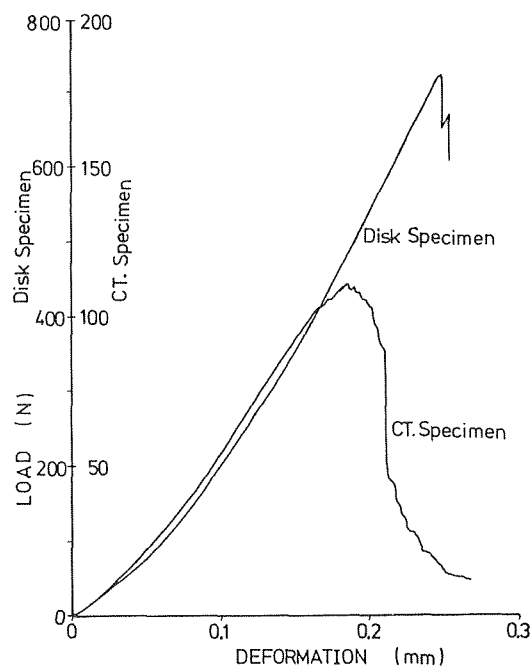


Fig. 7 Load-deformation curves in the disk test and the compact tensile test

shown in Fig. 7. In the compact tensile specimen, the curve of load-displacement appears irregular in part and final crack propagation occurs after some slow stable cracking. The ultimate load in the compact tensile test was used for calculation of the fracture toughness value K_{Ic} . Obtained results are $K_{Ic} = 0.817$ and $0.780 \text{ MPa m}^{1/2}$ for the disk and compact tensile specimens, respectively. The difference of these values is small. The deviation in the disk test was relatively smaller than that in the compact tensile test.

5.2 Mode I, II, and Combined Mode Fracture Toughness Testings.

The results of mode I, mode II, and combined mode I-II fracture toughness testing for some kinds of graphite, plaster and marble are shown as follows. Table 9 shows the results of reactor graphite 7477 as an example. Table 10 shows the summarized results of mode I and mode II fracture toughness and its deviations for all specimens. From this table, the ratios K_{IIc}/K_{Ic} are in the range of 1.1 ~ 1.2 independently of all materials. These values agree very well with recently investigated results of the combined mode fracture toughness testings for aluminum alloy and steel of which values K_{IIc}/K_{Ic} are in the range of 1.0 ~ 1.2 [6-9].

Fig. 8 shows the results of combined mode fracture toughness for all materials. This shows that the combined mode fracture criterion is also applicable in spite of the material difference. Representing the relationship of mode I-II combined mode fracture criterion as follows,

$$\left(\frac{K_I}{K_{Ic}}\right)^u + \left(\frac{K_{II}}{K_{IIc}}\right)^u = 1, \quad (21)$$

All the measured data are on the curve $u = 1.6$, except marble. The data of marble are on the circular curve $u = 2.0$ rather than on the foregoing curve. As a reason, it is considered that the

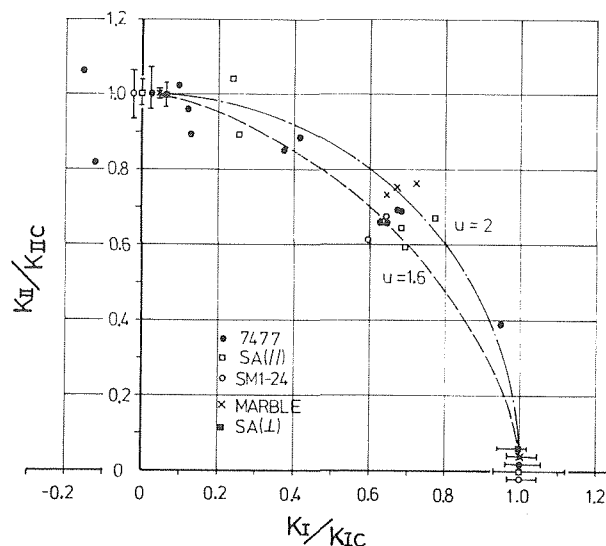


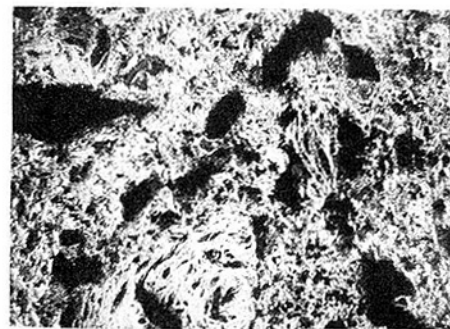
Fig. 8 Fracture criterion in the combined mode

Table 9 Results of mode I, mode II and combined mode fracture toughness of graphite 7477

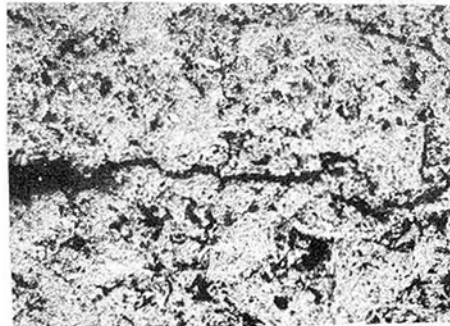
No.	c/R	θ (degree)	K_{Ic} or K_I (MPa $m^{1/2}$)	K_{IIc} or K_{II} (MPa $m^{1/2}$)
1	0.506	0	0.927	—
2	0.521	0	0.995	—
3	0.517	0	0.906	—
Ave.		0	0.943	—
4	0.500	22.9	—	1.05
5	0.599	20.2	—	1.05
6	0.489	23.2	—	1.10
7	0.466	23.7	—	1.17
Ave.			—	1.09
8	0.513	5	0.887	0.419
9	0.511	10	0.741	0.871
10	0.519	10	0.592	0.710
11	0.516	10	0.633	0.754
12	0.503	10	0.645	0.747
13	0.508	15	0.391	0.964
14	0.525	15	0.354	0.865
15	0.544	20	0.093	1.12
16	0.523	20	0.115	1.05
17	0.507	20	0.121	0.974
18	0.541	25	-0.115	0.893
19	0.538	25	-0.146	1.16

Table 10 Fracture toughness values of mode I and II

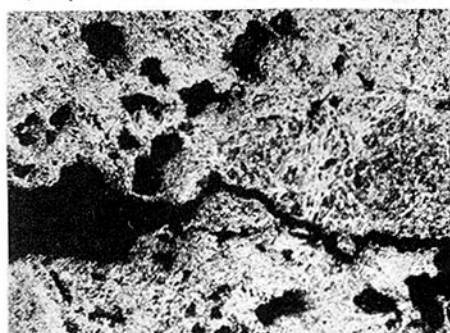
Materials	mode I			mode II			K_{IIc}/K_{Ic}
	K_{Ic} (MPa $m^{1/2}$)	Number of specimens	Standard deviation of K_{Ic}	K_{IIc} (MPa $m^{1/2}$)	Number of specimens	Standard deviation of K_{IIc}	
SA()	0.689	4	0.0203	0.762	5	0.0166	1.11
SA(//)	0.724	5	0.0586	0.830	3	0.0333	1.15
7477	0.943	3	0.0465	1.09	4	0.0568	1.16
SM1-24	0.814	3	0.0302	0.886	3	0.0575	1.09
Plaster	0.130	3	0.0060	0.148	4	0.0017	1.14
Marble	0.933	4	0.0303	1.05	3	0.0153	1.13



(a) SA(⊥) Mode II



(b) 7477 Mode I



(c) SM1-24 Mode I

Fig. 9 Crack propagation behavior

slit width of marble specimen, which is relatively larger than the other materials, influences the fracture toughness [24]. In any case, these are different from the result of Shah [7] which is $\alpha = 1.0$ as shown in equation (1).

Fig. 9 shows typical appearances of crack propagation from the tip of a slit in the disk tests of graphite. In the case of graphite, cracks meander through boundaries of filler cokes and link voids. In this figure, differences between the cracking appearance for mode I and mode II loading condition have been found insignificant.

The initial angles of crack propagation under mode II fracture were measured for graphites SA(L) and 7477. These values were almost 67° in average. But, in the case of graphite, the structures have many large pores, as seen in Fig. 9, and exact values are hard to obtain.

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